Olympics Sports Analyst Project

MECE Breakdown Report

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Project: Olympics Sports Data Analysis
Tools Used: MySQL, Microsoft Excel, Power BI

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1. Project Overview

The Olympics Sports Analyst project focuses on analyzing historical Olympic Games data to understand trends in sports, athlete participation, medal achievements, and country performance.

The main objective is to extract meaningful insights from Olympic data using SQL queries, Excel analysis, and Power BI visualizations.

2. Project Objectives

- Analyze the history and trends of Olympic Games.
- Understand the distribution of sports and events.
- Study athlete participation across different Olympic Games.
- Analyze medal achievements and performance.
- Identify countries and regions with strong Olympic participation and performance.

3. MECE Breakdown

The project is divided into five major dimensions to ensure comprehensive and structured analysis.

Dimension 1: Olympic Games and History

This dimension focuses on the historical structure and organization of Olympic Games.

Key areas:

- Summer and Winter Olympic Games.
- Olympic Games distribution across decades.
- Cities that hosted the highest number of Olympic Games.
- Changes in the number of events over time.

Business Questions:

- How many Summer and Winter Olympic Games have been conducted?
- How are Olympic Games distributed across decades?
- Which cities have hosted the most Olympic Games?
- How has the number of events changed over time?

Dimension 2: Sports and Events Analysis

This dimension focuses on the sports included in the Olympics and the development of events over time.

Key areas:

- Number of sports in the Olympics.
- Number of events in each sport.
- Sports distribution between Summer and Winter Games.
- Growth of sports participation over time.

Business Questions:

- Which sports have the highest number of events?
- How many events are conducted in each sport?
- How has participation in different sports evolved over time?
- How are events distributed across different gender categories?

Dimension 3: Athlete Participation Analysis

This dimension studies athlete participation and demographic patterns.

Key areas:

- Male and female athlete participation.
- Athlete participation across Olympic Games.
- Age distribution of athletes.
- Countries with the highest athlete participation.

Business Questions:

- How many athletes participated in the Olympics?
- What is the distribution of participants by gender?
- How does athlete age vary across Olympic Games?
- Which countries have the highest number of participants?

Dimension 4: Medal and Performance Analysis

This dimension focuses on medal achievements and Olympic performance.

Key areas:

- Medals awarded in each Olympic Games.
- Gold, Silver, and Bronze medal distribution.
- Countries with the highest gold medals.
- Medal distribution across different sports.

Business Questions:

- How many medals were awarded in each Olympic Games?
- Which countries have won the highest number of gold medals?
- How are medals distributed across different sports?
- Which regions have achieved the highest medal counts?

Dimension 5: Country and Regional Analysis

This dimension focuses on the geographical representation of Olympic participants.

Key areas:

- Number of participating regions/NOCs.
- Regional participation across Olympic Games.
- Regions with the highest athlete participation.
- Medal achievements by region.

Business Questions:

- How many regions participated in each Olympic Games?
- Which regions have the highest athlete participation?
- How has regional representation changed over time?
- How are medals distributed among different regions?

4. Tools and Technologies Used

Tool	Purpose
MySQL	Data querying and analysis
Microsoft Excel	Data analysis and charts
Power BI	Interactive dashboards and visualizations

5. Analytical Approach

The project follows these steps:

1. **Data Understanding:** Understanding the available tables and their relationships.
2. **Data Exploration:** Exploring Olympic datasets and identifying important fields.

3. **SQL Analysis:** Writing SQL queries to answer business questions.
4. **Excel Visualization:** Presenting query results through charts and tables.
5. **Power BI Dashboard:** Creating interactive dashboards to communicate insights.

6. Expected Business Value

This analysis helps users understand:

- Historical trends in Olympic Games.
- Growth and distribution of sports and events.
- Changes in athlete participation.
- Medal performance across countries and regions.
- Differences in participation and achievements over time.

7. Conclusion

The Olympics Sports Analyst project provides a structured analysis of historical Olympic data across five major dimensions: Olympic history, sports and events, athlete participation, medal performance, and regional representation.

By using MySQL, Excel, and Power BI, the project transforms raw data into meaningful insights and visual reports.

The MECE framework ensures that the analysis is organized, comprehensive, and easy to understand.